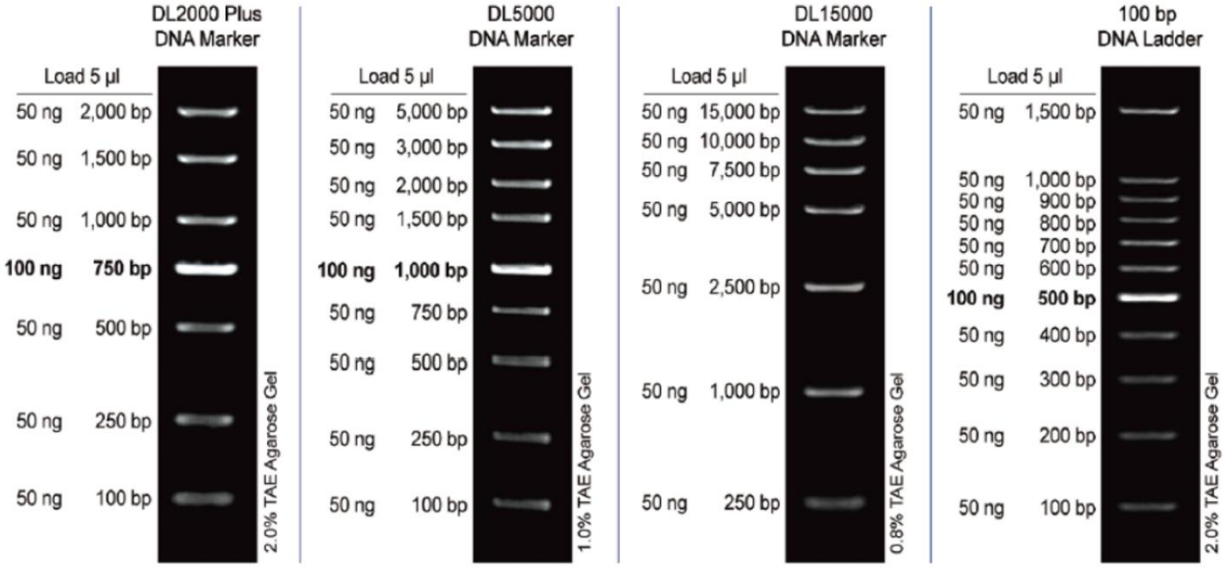
SCU China 2026 Notebook

Biosynthesis of β-ionone and β-citraurin

We used *Dunaliella salina* FACHB-435 as the primary chassis to investigate the biosynthesis of β-ionone and β-citraurin using CCD1 and CitCCD4, respectively. Using pCAMBIA1302 as the vector backbone, we compared expression constructs driven by the CaMV 35S promoter and the endogenous *D. salina* RBCS2 promoter. We also included a CCD1 construct carrying a chloroplast transit peptide. This notebook documents expression vector construction, *D. salina* transformation and selection, molecular identification, and expression verification.

Group ID	Corresponding part name	Abbreviation used in this notebook
C1	pRBCS2-CCD1	RBCS2-CCD1
C2	pRBCS2-CitCCD4	RBCS2-CitCCD4
C3	**pRBCS2-transit peptide-CCD1**	**RBCS2-SP-CCD1**
D1	p35S-CCD1	35S-CCD1
D2	p35S-CitCCD4	35S-CitCCD4

DNA ladder size reference



8.3

Plasmid backbone amplification and linearization by inverse PCR

We used pCAMBIA1302 as the template for inverse PCR to generate two types of linearized backbone: one retaining the 35S promoter and the other designed for replacement of the 35S promoter with the RBCS2 promoter.

25 µL reaction mixture for the RBCS2 group (four replicate reactions):

KOD One™ PCR Master Mix (2×)	12.5 µL
p1302-IF	2 µL
p1302-IR	2 µL
ddH_2O	7.5 µL
pCAMBIA1302	1 µL

25 µL reaction mixture for the 35S group (four replicate reactions):

KOD One™ PCR Master Mix (2×)	12.5 µL
p1302-IF	2 µL
p1302-35s-IR	2 µL
ddH_2O	7.5 µL
pCAMBIA1302	1 µL

Thermal cycling programme:

Initial denaturation	98 °C	10 min	
Denaturation	98 °C	10 s	34 cycles
Annealing	55 °C	5 s	
Extension	68 °C	2 min	
Final extension	68 °C	5 min	

Analysis of PCR products by agarose gel electrophoresis

PCR products were analysed on a 0.8% agarose gel stained with GelRed during gel preparation and imaged under UV illumination. A 250–15,000 bp DNA ladder was used.

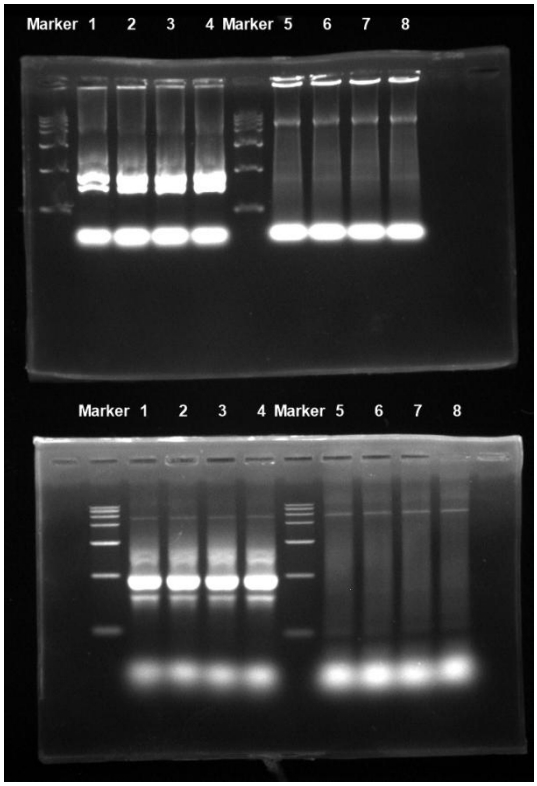


Fig. 1 | Agarose gel electrophoresis of plasmid backbone inverse PCR products.

Bands in the 35S group were faint, whereas the RBCS2 group produced prominent non-specific bands at approximately 1,000 bp. Neither group yielded satisfactory backbone amplification. We therefore planned to adjust the PCR conditions and repeat the amplification.

8.4

Using revised reaction mixtures and thermal cycling conditions, we repeated inverse PCR with pCAMBIA1302 as the template to generate a backbone retaining the 35S promoter and a backbone for replacement with the RBCS2 promoter.

Plasmid backbone linearization by inverse PCR

Revised reaction mixture: primer concentrations were reduced because of the large plasmid size.

KOD One™ PCR Master Mix (2×)	50 µL
p1302-IF	1.5 µL
p1302-IR	1.5 µL
ddH_2O	46 µL
pCAMBIA1302	1 µL

Thermal cycling programme for comparison of annealing temperatures:

Initial denaturation	98 °C	3 min	
Denaturation	98 °C	10 s	34 cycles
Annealing	64, 62, 60, 58 °C	5 s	
Extension	68 °C	2 min	
Final extension	68 °C	5 min	

Touchdown PCR programme:

Initial denaturation	98 °C	3 min	
Denaturation	98 °C	10 s	16 cycles
Annealing	66–58 °C (decreased by 0.5 °C per cycle)	5 s	
Extension	68 °C	2 min	
Denaturation	98 °C	10 s	18 cycles
Annealing	58 °C	5 s	
Extension	68 °C	2 min	
Final extension	68 °C	5 min	

Analysis of PCR products by agarose gel electrophoresis

PCR products were analysed on a 0.8% agarose gel stained with GelRed during gel preparation and imaged under UV illumination. A 250–15,000 bp DNA ladder was used.

Touchdown PCR results:

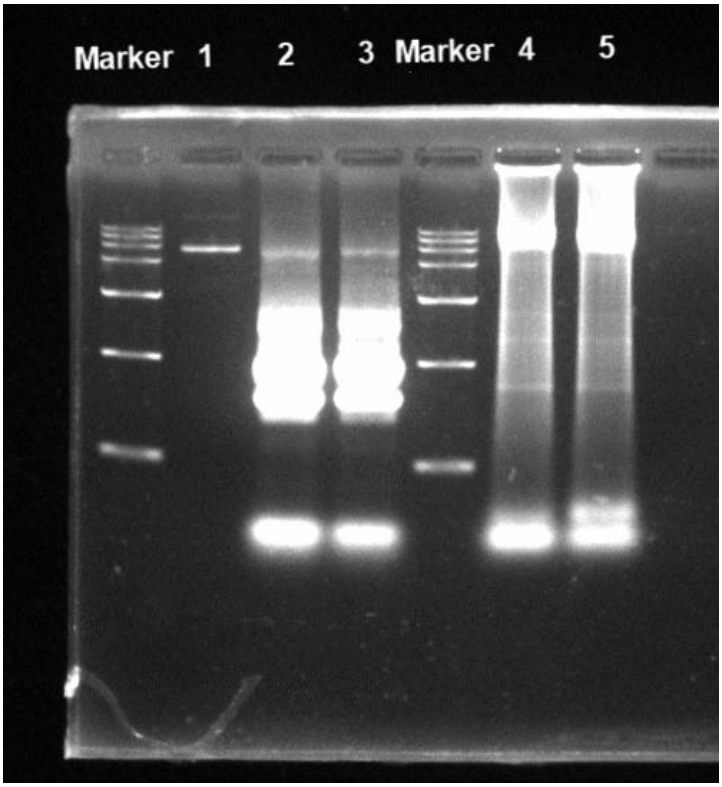


Fig. 2 | Agarose gel electrophoresis of touchdown PCR products. Lanes from left to right: DNA ladder, circular plasmid control, RBCS2 group, RBCS2 group, DNA ladder, 35S group, and 35S group.

Results at different annealing temperatures:

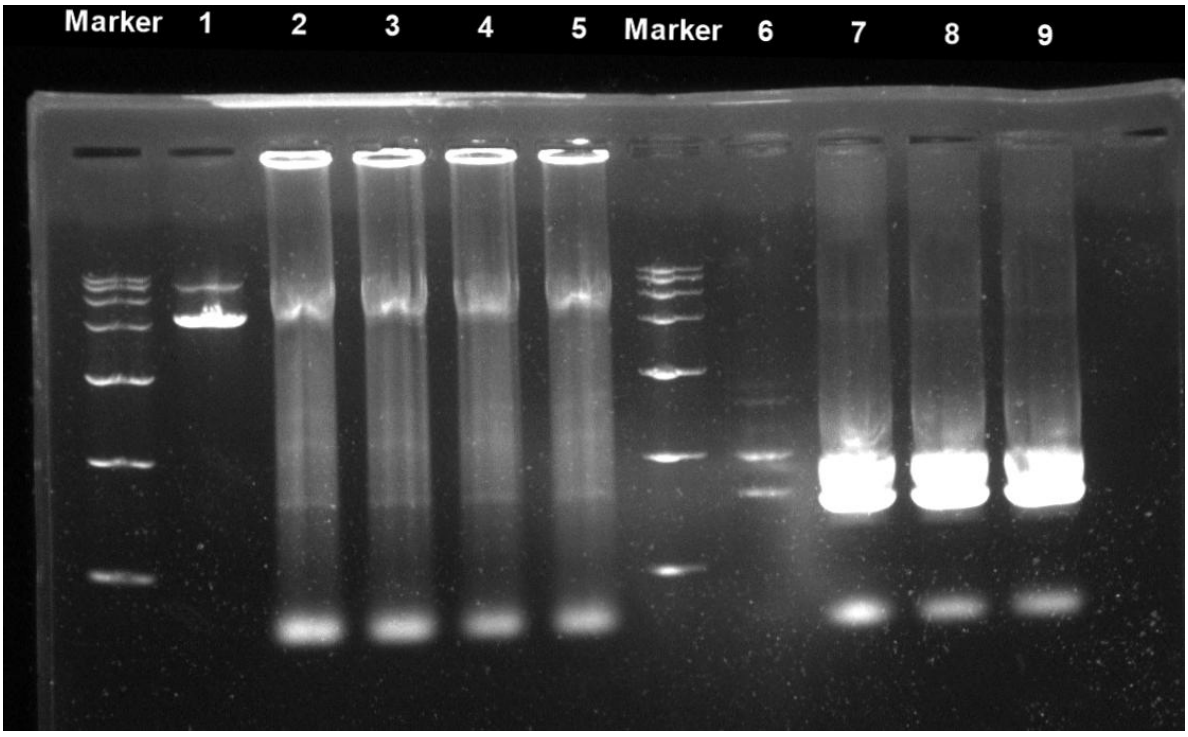


Fig. 3 | Agarose gel electrophoresis of PCR products obtained at different annealing temperatures. Lanes from left to right: DNA ladder; circular plasmid control; 35S group at 58, 60, 62, and 64 °C; DNA ladder; and RBCS2 group at 58, 60, 62, and 64 °C.

The 35S group produced more distinct bands. The RBCS2 group showed the same problem as on the previous day, with prominent non-specific bands at approximately 1,000 bp. We suspected that the primers annealed to additional sites and redesigned the primers for the RBCS2 group.

8.5

Plasmid backbone amplification and linearization by inverse PCR

We continued attempts to linearize pCAMBIA1302 by inverse PCR, replacing the PCR premix with Phanta Ultra Master Mix.

25 µL reaction mixture for the RBCS2 group:

2× Phanta Ultra Master Mix (Dye Plus)	12.5 µL
p1302-IF	0.75 µL
p1302-IR	0.75 µL
ddH_2O	10.75 µL
pCAMBIA1302	0.25 µL

25 µL reaction mixture for the 35S group:

2× Phanta Ultra Master Mix (Dye Plus)	12.5 µL
p1302-IF	0.75 µL
p1302-35s-IR	0.75 µL
ddH_2O	10.75 µL
pCAMBIA1302	0.25 µL

PCR programme:

Initial denaturation	98 °C	3 min	
Denaturation	98 °C	10 s	30 cycles
Annealing	60 °C	5 s	
Extension	68 °C	1 min 30 s	
Final extension	68 °C	5 min	

Analysis of PCR products by agarose gel electrophoresis

PCR products were analysed on a 0.8% agarose gel stained with GelRed during gel preparation and imaged under UV illumination. A 250–15,000 bp DNA ladder was used.

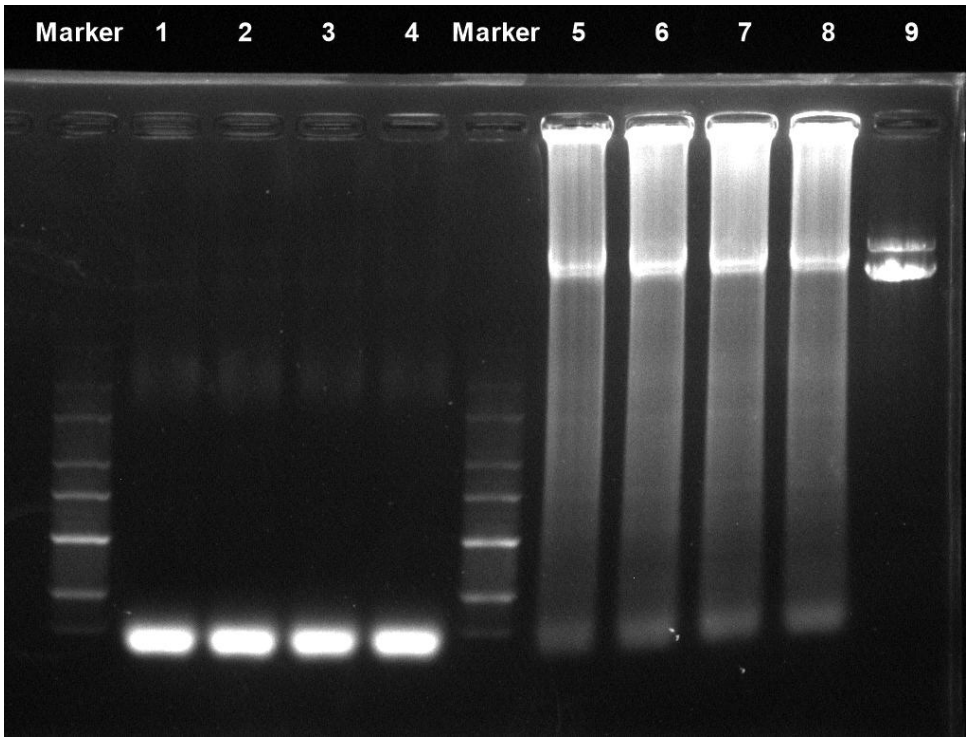


Fig. 4 | Agarose gel electrophoresis of PCR products.

Expansion of plasmid-bearing cultures and recovery of strains carrying target genes

Working in a clean bench, we picked a single *Escherichia coli* colony carrying pCAMBIA1302 from an agar plate to establish an expanded culture.

We recovered glycerol stocks of strains carrying CCD1 or CitCCD4 to prepare templates for subsequent plasmid extraction and amplification of the target fragments.

8.6

Plasmid extraction

Plasmids were extracted using a TIANGEN plasmid miniprep kit (DP103) according to the manufacturer's instructions.

We performed plasmid minipreps on the DH5α-p1302 culture and the strains carrying the target-gene templates expanded on the previous day. Two p1302 plasmid preparations were obtained at concentrations of 267.60 and 104.10 ng/μL.

Plasmids were extracted from the recovered CCD1 and CitCCD4 glycerol stock cultures. The CitCCD4 template plasmid concentration was 179.00 ng/μL. The two CCD1 template plasmid preparations had concentrations of 142.30 and 37.750 ng/μL, and the RBCS2+SP template plasmid concentration was 222.05 ng/μL.

Plasmid backbone amplification and linearization by inverse PCR

25 µL reaction mixture for the RBCS2 group (four replicate reactions):

2× Phanta Ultra Master Mix (Dye Plus)	12.5 µL
p1302-IF	0.75 µL
p1302-IR	0.75 µL
ddH_2O	10.75 µL
pCAMBIA1302	0.25 µL

PCR programme:

Initial denaturation	98 °C	3 min	
Denaturation	98 °C	10 s	30 cycles
Annealing	60 °C	5 s	
Extension	68 °C	1 min 30 s	
Final extension	68 °C	5 min	

Agarose gel electrophoresis and gel extraction

PCR products were analysed on a 0.8% agarose gel stained with GelRed during gel preparation and imaged under UV illumination. A 250–15,000 bp DNA ladder was used.

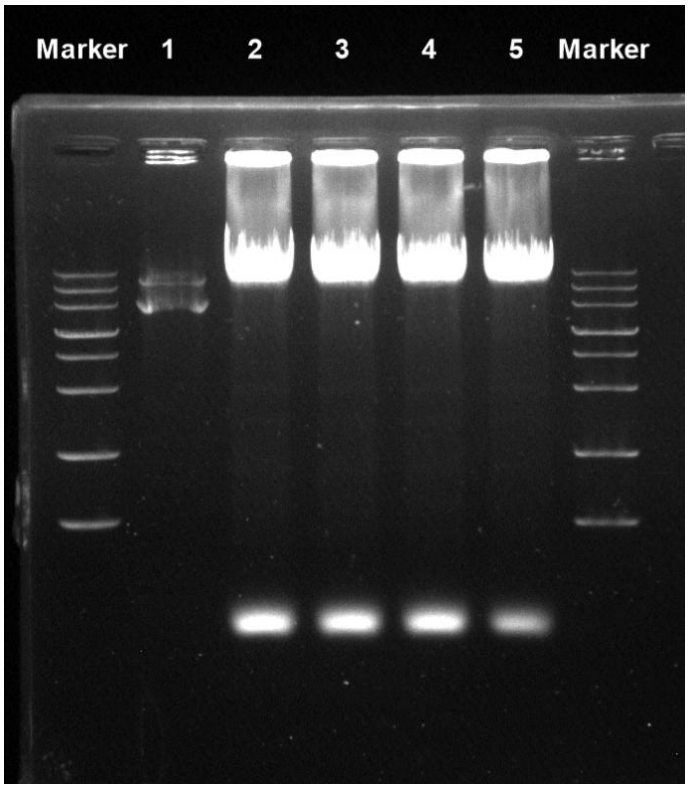


Fig. 5 | Agarose gel electrophoresis of PCR products.

DNA was recovered using an Omega E.Z.N.A.® Gel Extraction Kit according to the manufacturer's instructions.

The recovered backbone concentrations were 63.000 ng/µL for the RBCS2 group and 13.250 ng/µL for the 35S group. Given the low concentration in the 35S group, we suspected that migration of the large DNA fragment through the gel might have been affected and considered reducing the agarose concentration.

8.7

Amplification of target fragments with homology arms

We amplified insert fragments containing homology arms from the CCD1, CitCCD4, and RBCS2 template constructs.

Reaction mixture for C1 CCD1:

KOD One™ PCR Master Mix (2×)	12.5 μL
C1-CCD1-F	0.75 μL
CCD1-R	0.75 μL
ddH_2O	10.5 μL
pCAMBIA1302	0.5 μL

Reaction mixture for C3 CCD1:

KOD One™ PCR Master Mix (2×)	12.5 μL
C3-CCD1noATG-F	0.75 μL
CCD1-R	0.75 μL
ddH_2O	10.5 μL
pCAMBIA1302	0.5 μL

Reaction mixture for C2 CitCCD4:

KOD One™ PCR Master Mix (2×)	12.5 μL
C2-CitCCD4-F	0.75 μL
CitCCD4-R	0.75 μL
ddH_2O	10.5 μL
pCAMBIA1302	0.5 μL

Reaction mixture for C1 RBCS2:

KOD One™ PCR Master Mix (2×)	12.5 μL
RBCS2-F	0.75 μL
C1-RBCS2-R	0.75 μL
ddH_2O	10.5 μL
pCAMBIA1302	0.5 μL

Reaction mixture for D1 CCD1:

KOD One™ PCR Master Mix (2×)	12.5 μL
D1-CCD1-F	0.75 μL
CCD1-R	0.75 μL
ddH_2O	10.5 μL
pCAMBIA1302	0.5 μL

Reaction mixture for D2 CitCCD4:

KOD One™ PCR Master Mix (2×)	12.5 μL
D2-CitCCD4-F	0.75 μL
CitCCD4-R	0.75 μL
ddH_2O	10.5 μL
pCAMBIA1302	0.5 μL

Reaction mixture for C3 RBCS2:

KOD One™ PCR Master Mix (2×)	12.5 μL
RBCS2-F	0.75 μL
C3-RBCS2SP-R	0.75 μL
ddH_2O	10.5 μL
pCAMBIA1302	0.5 μL

Reaction mixture for C2 RBCS2:

KOD One™ PCR Master Mix (2×)	12.5 μL
RBCS2-F	0.75 μL
C2-RBCS2-R	0.75 μL
ddH_2O	10.5 μL
pCAMBIA1302	0.5 μL

PCR programme for D2 CitCCD4:

Initial denaturation	98 °C	3 min	
Denaturation	98 °C	10 s	34 cycles
Annealing	55 °C	5 s	
Extension	68 °C	20 s	
Final extension	68 °C	3 min	

PCR programme for all groups other than D2 CitCCD4:

Initial denaturation	98 °C	3 min	
Denaturation	98 °C	10 s	34 cycles
Annealing	60 °C	5 s	
Extension	68 °C	20 s	
Final extension	68 °C	3 min	

Agarose gel electrophoresis and gel extraction

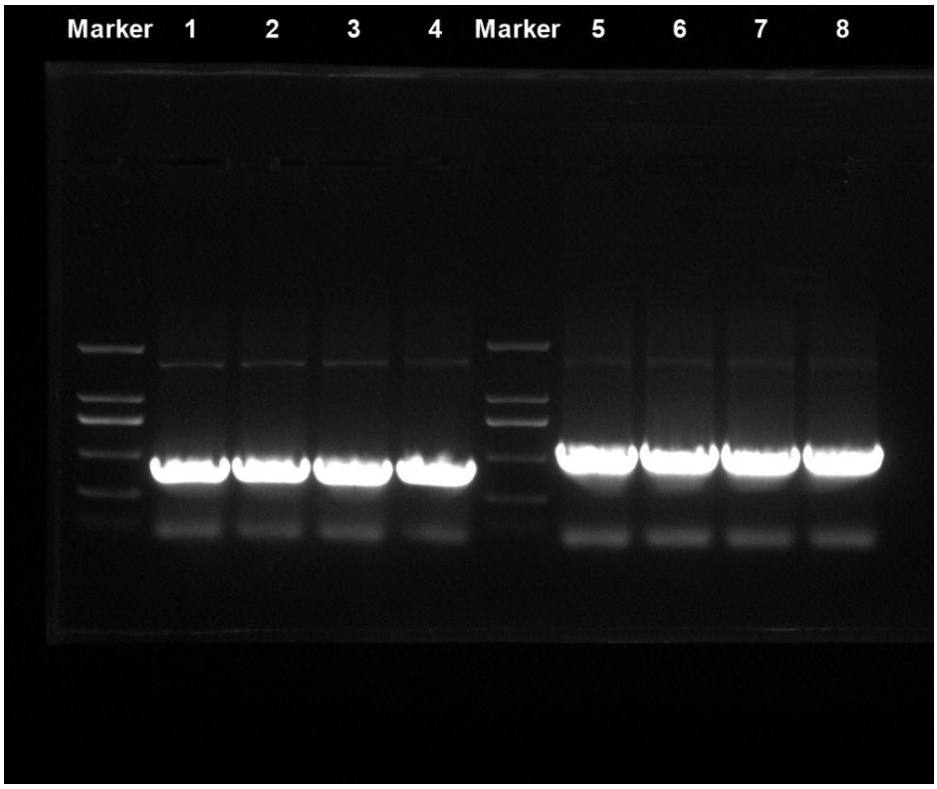


Fig. 6 | Agarose gel electrophoresis of PCR products. Lanes 1–4, C2 RBCS2; lanes 5–8, C3 RBCS2. A 100–2,000 bp DNA ladder was used.

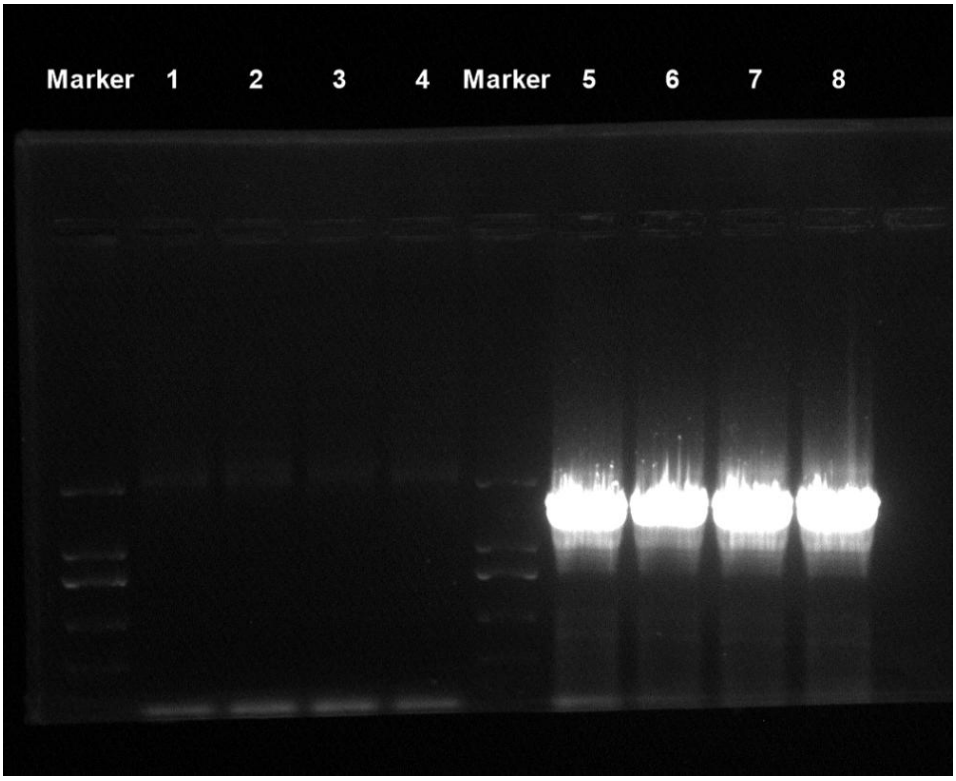


Fig. 7 | Agarose gel electrophoresis of PCR products. Lanes 1–4, C3 CCD1; lanes 5–8, C2 CitCCD4. A 100–2,000 bp DNA ladder was used.

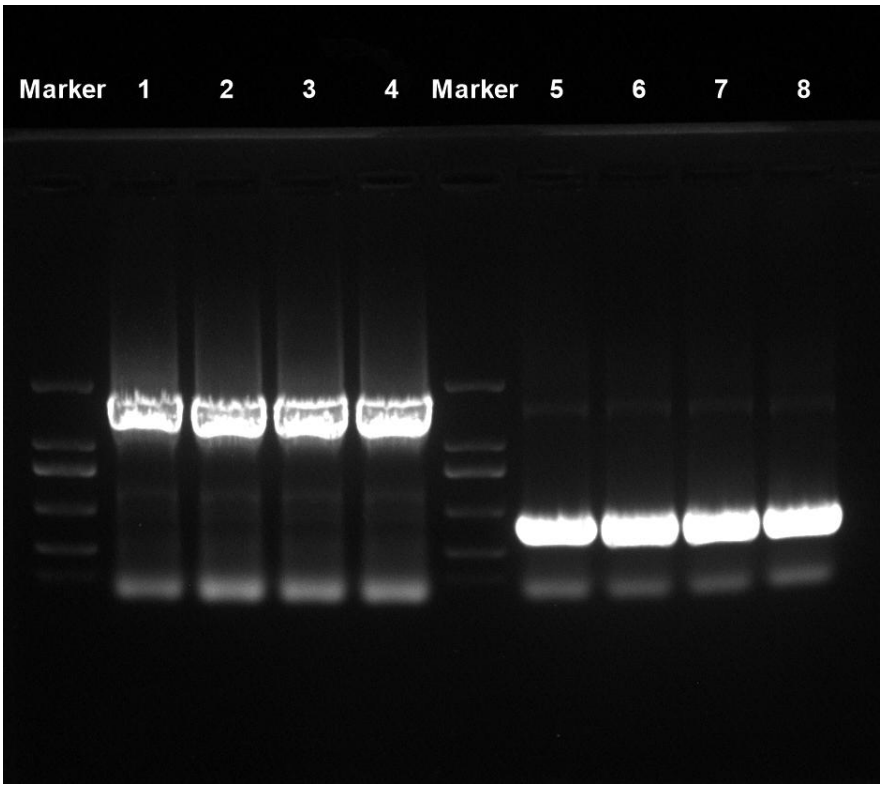


Fig. 8 | Agarose gel electrophoresis of PCR products. Lanes 1–4, D2 CitCCD4; lanes 5–8, C1 RBCS2. A 100–2,000 bp DNA ladder was used.

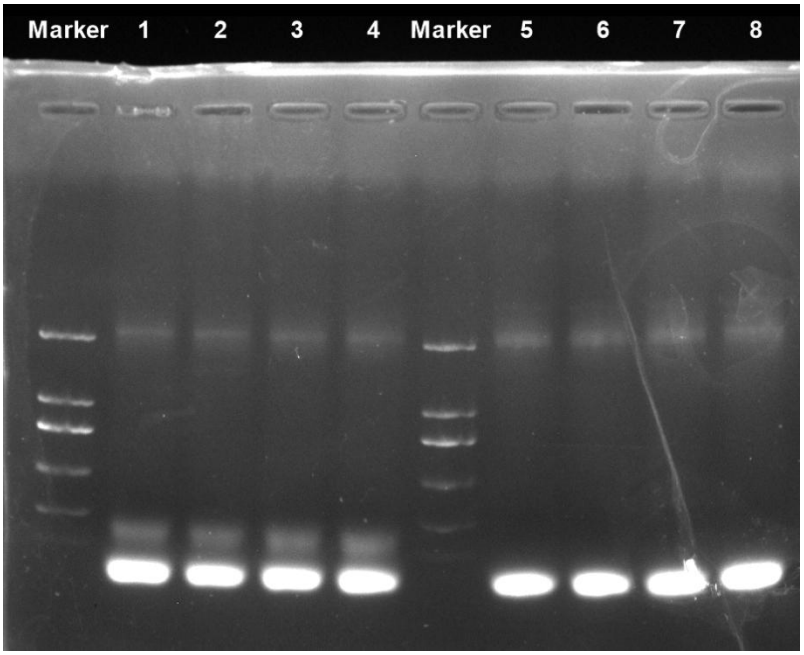


Fig. 9 | Agarose gel electrophoresis of PCR products. Lanes 1–4, D1 CCD1; lanes 5–8, C1 CCD1. A 100–2,000 bp DNA ladder was used.

PCR products were analysed on a 1.0% agarose gel stained with GelRed during gel preparation and imaged under UV illumination. A 100–2,000 bp DNA ladder was used. The CCD1 fragment bands were extremely faint, whereas the CitCCD4 and RBCS2 fragments amplified more efficiently. The concentrations obtained after gel extraction using the Omega kit are listed below. The CCD1 fragments required re-amplification.

Fragment	Concentration after gel extraction (ng/µL)
C1 CCD1	2.15
C3 CCD1	1.75
D1 CCD1	1.25
C2 CitCCD4	88.00
D2 CitCCD4	90.90
C1 RBCS2+SP	217.95
C2 RBCS2+SP	208.20
C3 RBCS2+SP	164.65

8.8

Optimization of CCD1 fragment amplification with homology arms

We re-amplified the CCD1 fragments, comparing annealing temperatures of 45 and 65 °C using the reaction mixtures and programme described on the previous day. The gel electrophoresis results are shown below. A 100–2,000 bp DNA ladder was used.

At an annealing temperature of 45 °C, numerous non-specific products were observed and the target bands were indistinct:

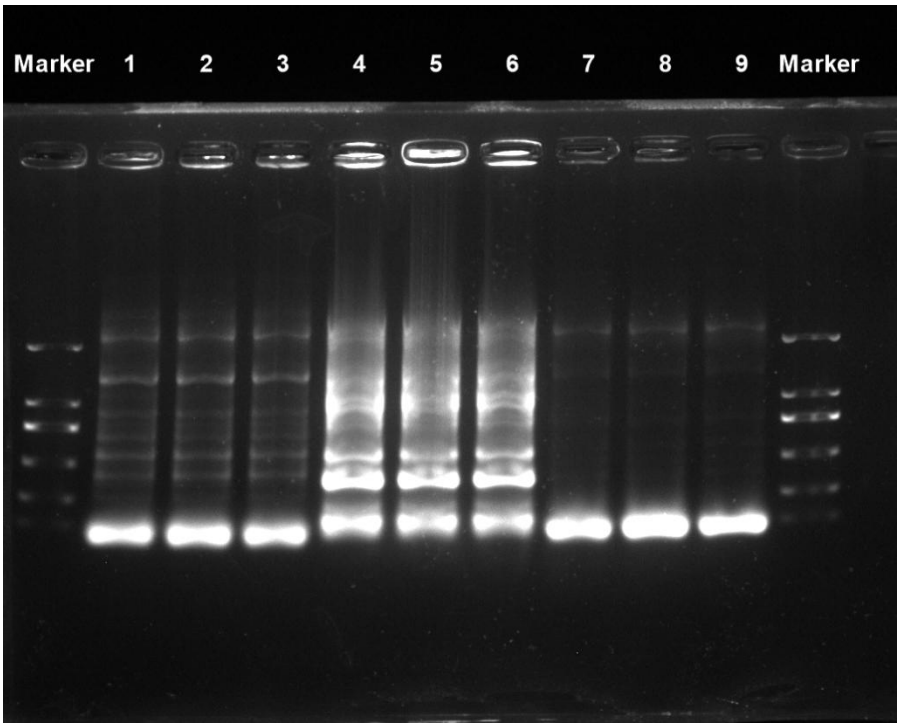


Fig. 10 | Agarose gel electrophoresis of PCR products amplified at an annealing temperature of 45 °C. Lanes 1–3, 4–6, and 6–9 correspond to C1 CCD1, C3 CCD1, and D1 CCD1, respectively.

At an annealing temperature of 65 °C, fewer non-specific products were observed, but the target DNA concentration was insufficient for homology-based assembly:

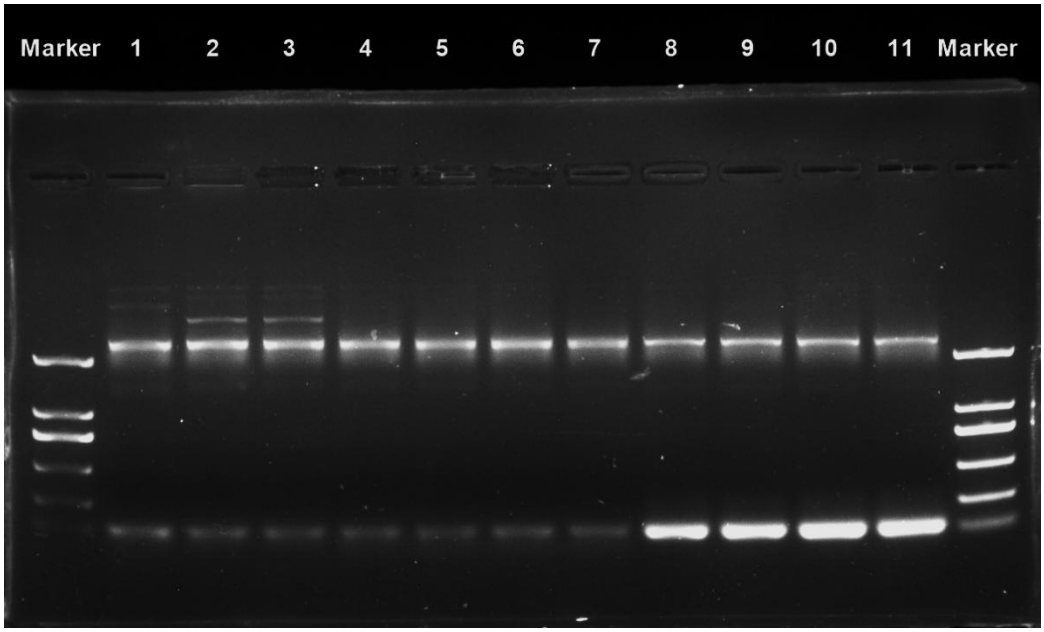


Fig. 11 | Agarose gel electrophoresis of PCR products amplified at an annealing temperature of 65 °C. Lanes 1–3, 4–7, and 8–11 correspond to C1 CCD1, C3 CCD1, and D1 CCD1, respectively.

Homology-based assembly of CitCCD4 constructs and subsequent experiments

Using the available backbones and CitCCD4 insert fragments, we first assembled the D2 and C2 vectors.

10 µL Gibson assembly reaction mixture for D2 to generate p1302-35S-CitCCD4:

Component	Volume
2× CloneExpress ②	5 µL
35S backbone	4 µL
D2 CitCCD4 (90.9)	1 µL

10 µL Gibson assembly reaction mixture for C2 to generate p1302-RBCS2-CitCCD4:

Component	Volume
2× CloneExpress ②	5 µL
RBCS2 backbone (63)	2.95 µL
C2 RBCS2+SP (208.20)	0.11 µL
C2 CitCCD4 (88)	0.78 µL
ddH_2O	1.16 µL

Assembly conditions: incubate at 50 °C for 30 min, then use immediately or store at −20 °C.

Transformation by heat shock: (1) Thaw 100 µL competent cells on ice, add the target DNA, and mix gently on ice. (2) Heat-shock the cells in a 42 °C water bath for 45–60 s, then immediately place them on ice for 2 min. (3) Add 700 µL antibiotic-free liquid LB medium and allow recovery at 37 °C and 200 rpm for 60 min. (4) Plate the required volume on kanamycin-containing LB agar and incubate overnight at 37 °C.

8.11

Preliminary screening of CitCCD4 clones

CitCCD4 clones were screened by PCR using bacterial cultures as templates, and the remaining cultures were expanded.

Reaction mixture for D2 CitCCD4:

KOD One™ PCR Master Mix (2×)	12.5 µL
D2-CitCCD4-F	0.75 µL
CitCCD4-R	0.75 µL
ddH_2O	10 µL
Bacterial culture	1 µL

Reaction mixture for C2 CitCCD4:

KOD One™ PCR Master Mix (2×)	12.5 µL
C2-RBCS2-F	0.75 µL
CitCCD4-R	0.75 µL
ddH_2O	10 µL
Bacterial culture	1 µL

PCR cycling conditions:

Initial denaturation	98 °C	3 min	
Denaturation	98 °C	10 s	34 cycles
Annealing	65 °C	5 s	
Extension	68 °C	20 s	
Final extension	68 °C	3 min	

PCR products were analysed by electrophoresis on a 1% agarose gel, as shown below. A 100–2,000 bp DNA ladder was used.

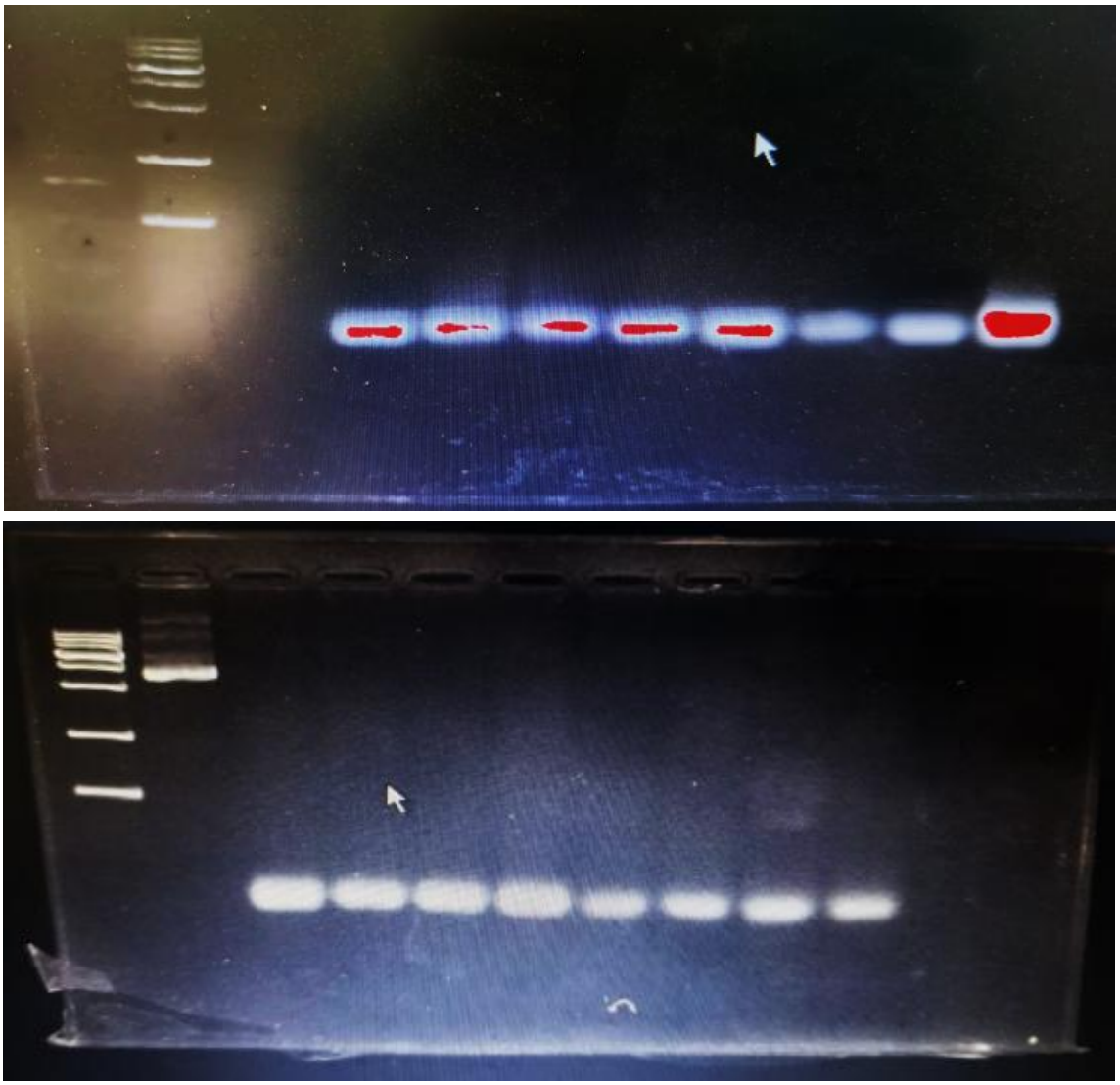


Fig. 12 | Agarose gel electrophoresis of PCR products from the CitCCD4 constructs. The two DNA ladders are shown on the left.

No positive PCR results were obtained for the CitCCD4 constructs on this day.

Homology-based assembly of CCD1 constructs and subsequent procedures



Fig. 13 | Agarose gel electrophoresis of PCR products for the CCD1 constructs. Images show CCD1-C1, CCD1-C3 and CCD1-D1, respectively. A 100–2,000 bp DNA ladder was used.

Electrophoresis on a 1% agarose gel yielded bands corresponding to the target CCD1 fragments, represented by the larger DNA bands.

The target fragments were gel-purified, yielding two preparations each for C1, C3 and D1. Their concentrations are listed below. The CCD1 constructs were subsequently assembled by homology-based assembly, introduced into DH5α cells and plated for cultivation. Bacterial stocks were also prepared.

Sample	Concentration of preparation 1 (ng/μL)	Concentration of preparation 2 (ng/μL)
C1 CCD1	263.10	257.70
C3 CCD1	155.60	187.55
D1 CCD1	239.10	256.00

Homology-based assembly was used to construct p1302-35S-CCD1, p1302-RBCS2-CCD1 and p1302-RBCS2-SP-CCD1. Reaction mixtures were calculated using the previous day's conditions and the manufacturer's instructions. Reactions were incubated at 50 °C for 30 min.

The recombinant CCD1 plasmids (C1, C3 and D1) were introduced into DH5α cells by heat shock and plated for cultivation. (1) An aliquot of 100 μL competent cells was thawed on ice, mixed gently with the target DNA and kept on ice. (2) Cells were heat-shocked in a 42 °C water bath for 45–60 s and immediately placed on ice for 2 min. (3) A volume of 700 μL antibiotic-free liquid LB medium was added, and cells were allowed to recover at 37 °C and 200 rpm for 60 min. (4) Appropriate volumes were spread onto kanamycin-containing LB plates and incubated overnight at 37 °C.

8.12

Screening of CitCCD4 clones

PCR screening of bacterial cultures was repeated with adjusted cycling conditions. The reaction mixtures were unchanged from the previous day.

PCR cycling conditions:

Initial denaturation	98 °C	3 min	
Denaturation	98 °C	10 s	34 cycles
Annealing	65 °C	5 s	
Extension	68 °C	30 s	
Final extension	68 °C	3 min	

Plasmid extraction, culture expansion and stock preparation for PCR-positive clones

PCR products were assessed by electrophoresis on a 1% agarose gel, as shown below.

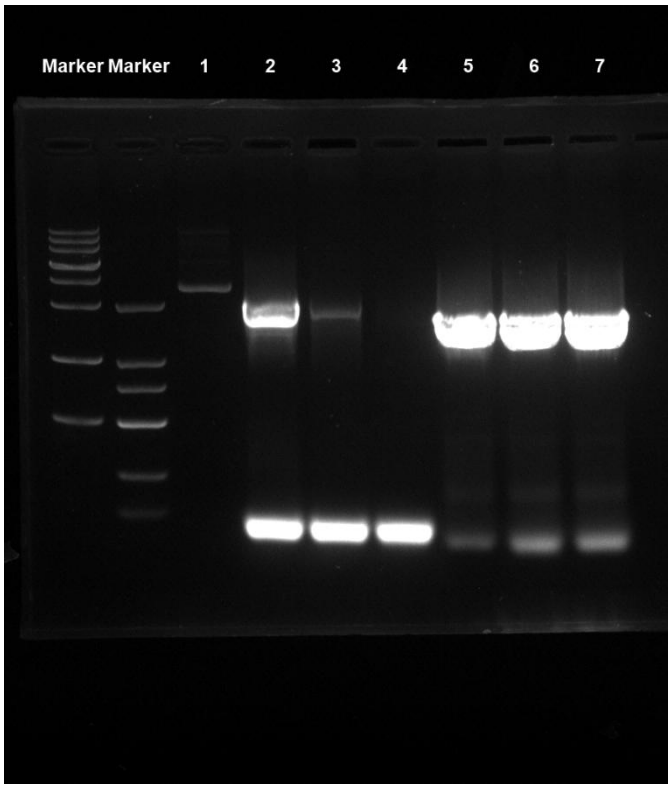


Fig. 14 | Agarose gel electrophoresis of PCR products from bacterial cultures carrying the CitCCD4 constructs. The left and right DNA ladders span 250–15,000 bp and 100–2,000 bp, respectively. Lanes 2–4, C2; lanes 5–7, D2.

Clear positive bands were observed in the first C2 lane and all three D2 lanes. Plasmids were extracted from the corresponding clones, and the cultures were expanded and preserved as bacterial stocks.

Plasmid extraction results:

Clone ID	Concentration (ng/μL)
c1-1	62.25
d1-1	32.00
d1-2	32.45
d1-3	38.5

Preparation of CCD1 clones

Single colonies were picked from the previous day's CCD1 transformation plates and expanded for subsequent colony PCR and recombinant plasmid extraction.

8.13

Screening of CCD1 clones and plasmid extraction

The expanded CCD1 cultures were screened by PCR, and the CitCCD4 cultures were examined in parallel.

Reaction mixture for C1:

KOD One™ PCR Master Mix (2×)	12.5 μL
C1-CCD1-F	0.75 μL
CCD1-R	0.75 μL
ddH_2O	10 μL
Bacterial culture	1 μL

Reaction mixture for C3:

KOD One™ PCR Master Mix (2×)	12.5 μL
C3-CCD1-F	0.75 μL
CCD1-R	0.75 μL
ddH_2O	10 μL
Bacterial culture	1 μL

Reaction mixture for D1:

KOD One™ PCR Master Mix (2×)	12.5 μL
D1-CCD1-F	0.75 μL
CCD1-R	0.75 μL
ddH_2O	10 μL
Bacterial culture	1 μL

Reaction mixture for D2 CitCCD4:

KOD One™ PCR Master Mix (2×)	12.5 μL
D2-CitCCD4-F	0.75 μL
CitCCD4-R	0.75 μL
ddH_2O	10 μL
Bacterial culture	1 μL

Reaction mixture for C2 CitCCD4:

KOD One™ PCR Master Mix (2×)	12.5 μL
C2-RBCS2-F	0.75 μL
CitCCD4-R	0.75 μL
ddH_2O	10 μL
Bacterial culture	1 μL

Following agarose gel electrophoresis, plasmids were extracted from selected cultures.

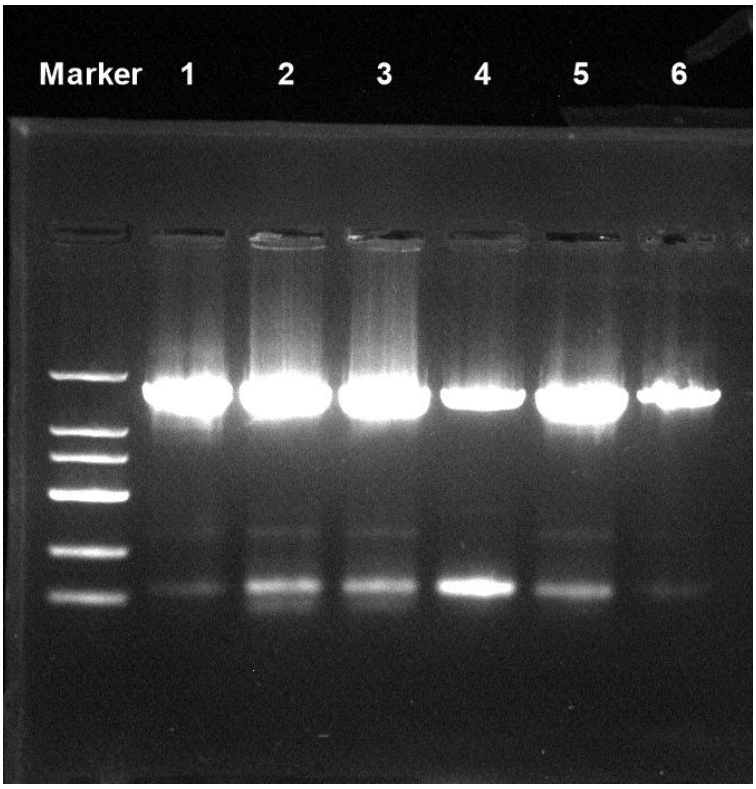


Fig. 15 | Agarose gel electrophoresis of PCR products from bacterial cultures carrying the CCD1 constructs. Lanes 1–6, D1. A 100–2,000 bp DNA ladder was used.

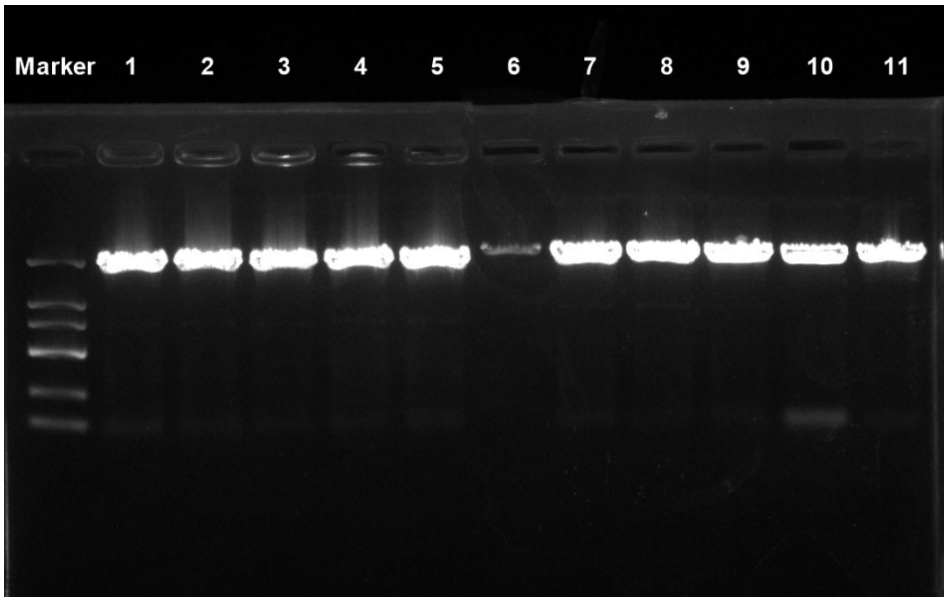


Fig. 16 | Agarose gel electrophoresis of PCR products from bacterial cultures carrying the CCD1 constructs. Lanes 1–5, C1; lanes 7–11, C3. A 100–2,000 bp DNA ladder was used.

Plasmid extraction results are listed below.

Clone ID	Concentration (ng/μL)
RBCS2-1	117
35s-1	210
35s-2	170
35s-3	120
C1-1	<50
C1-5	<50

Clone ID	Concentration (ng/μL)
C1-6	82
C3-2	56
C3-3	<50
C3-4	<50
D1-2	>100
D1-3	>100
D1-5	>100

8.14

Sequencing of recombinant CCD1 plasmids

The extracted recombinant CCD1 plasmids were submitted for sequencing. All recombinant bacterial strains were expanded again by inoculating 250 μL of each original culture into 5 mL of fresh antibiotic-free liquid LB medium and incubating overnight at 37 °C and 200 rpm.

Preparation for *Dunaliella salina* cultivation

Preparation of DM medium containing 1.5 M NaCl.

The medium composition per litre is listed below. Medium was dispensed into Erlenmeyer flasks, each containing 50 mL of medium and 5 mL of an existing high-density *D. salina* culture, to expand the algal cultures.

Component	Amount per litre
NaCl	87.69 g
$NaNO_3$	0.42 g
$NaH_2PO_4 \cdot 2H_2O$	15.60 mg
$MgSO_4 \cdot 7H_2O$	1.23 g
KCl	74.00 mg
$CaCl_2 \cdot 2H_2O$	44.00 mg
10× A5 solution	100 μL
$NaHCO_3$	0.84 g
Fe·EDTA solution	1 mL

8.17

Plasmid extraction

Based on the previous sequencing results, plasmids were extracted from 10 bacterial clones: D2-1, D2-3, D1-2, D1-3, C1-5, C1-6, C2, C3-2, C3-3 and C3-4. Bacterial stocks were also prepared. These plasmids were then used for the first batch of *D. salina* transformations.

Transformation of *Dunaliella salina* by osmotic shock

Using the procedure below, two tubes were prepared for each construct from C1 to D2, giving 10 tubes of transformed algal cultures in total.

Cell cultivation and pretreatment: *D. salina* FACHB-435 was grown without agitation in modified liquid PKS/DM medium containing 1.5 M NaCl at 26 °C, under a photon flux density of 50 μmol photons m^{-2} s^{-1} and a 16 h light period per day. Cultures were grown to the logarithmic phase, corresponding to approximately 10^5–10^6 cells/mL and an OD_{680} of approximately 0.5–1.0.

1. Harvest cells from a logarithmic-phase culture by centrifugation.

2. Discard the supernatant, resuspend the pellet in fresh PKS/DM medium and wash the cells three times to remove residual metabolites from the spent medium.

3. Resuspend the washed cells in fresh medium and adjust the final cell density to 10^6 cells/mL.

4. Transfer the suspension to fresh medium containing 1 M NaCl and incubate for a further 8 h to acclimate the cells before osmotic shock.

5. Collect 1 mL of the pretreated cell suspension by centrifugation and discard the supernatant.

6. Immediately resuspend the pellet in fresh PKS/DM medium containing 0.1 M NaCl, maintaining a volume of 1 mL.

7. Add the following to the 1 mL suspension in the specified order:

Plasmid DNA: 5.46 μg of pCAMBIA1302.

Surfactant: 15 μL of 0.1% Triton X-100 stock solution to facilitate membrane permeabilization.

8. Vortex the mixture vigorously for 60–120 s.

9. Immediately after vortexing, transfer the mixture into 30 mL of fresh liquid medium.

8.18

Cultivation of transformed algae

The first batch of transformed *D. salina* cultures was maintained at 26 °C with a 16 h light period per day.

Expansion of bacterial cultures

Escherichia coli carrying the recombinant plasmids was expanded in kanamycin-containing liquid LB medium to provide biomass for subsequent larger-scale plasmid extraction.

8.19

Screening across a hygromycin concentration gradient at 1.5 M NaCl

Hygromycin concentration (mg/L)	0	200	500	1000	1500

8.20

Algal growth 48 h after transformation

Growth of the previously transformed algal cultures was examined. The cultures were only faintly green, and centrifugation yielded a small green pellet, indicating the presence of algal cells. The cells were resuspended and cultivation was continued; no obvious increase in growth was visible after a further 24 h.

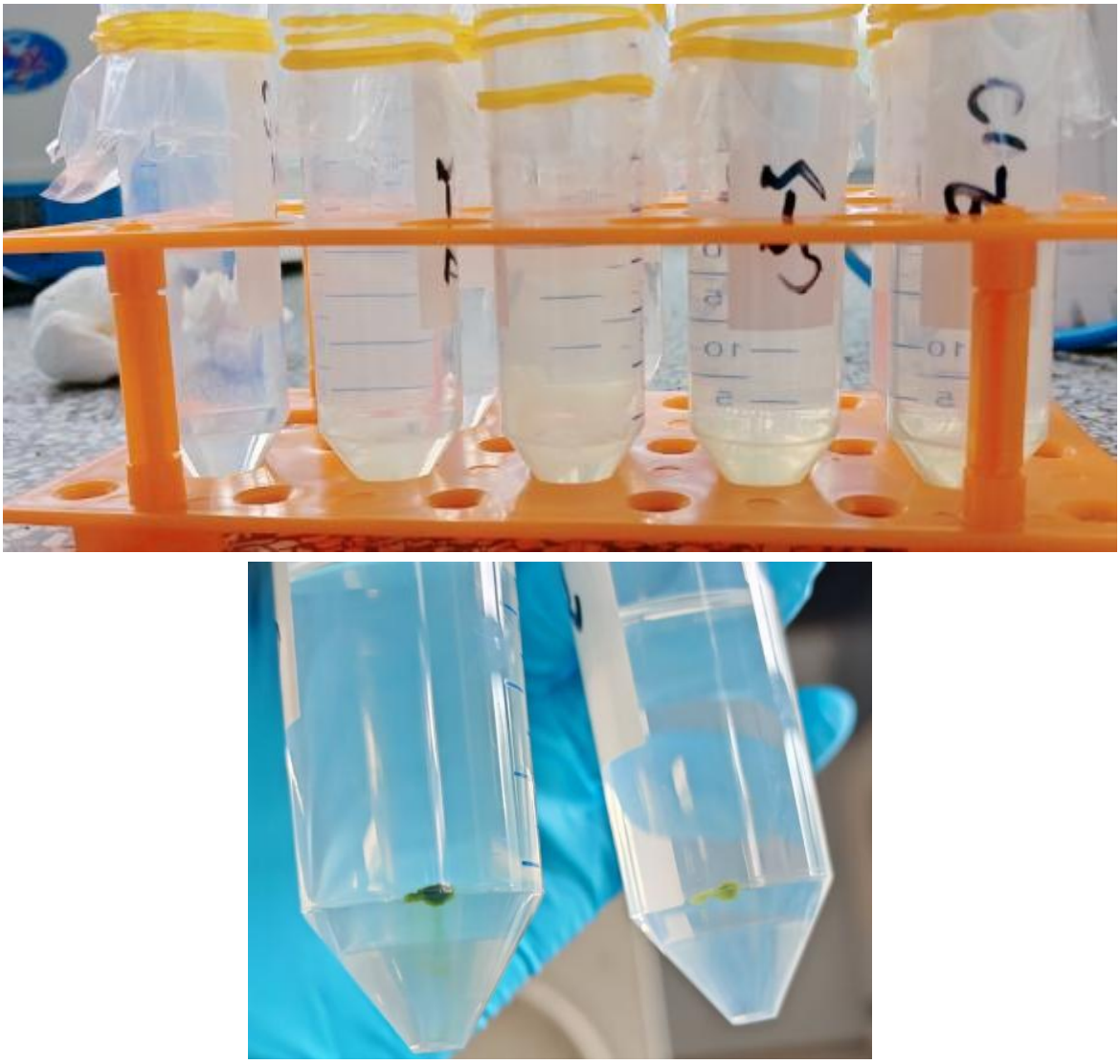


Fig. 17 | Growth of transformed *Dunaliella salina* 48 h after transformation.

Plasmid extraction from bacterial cultures

In parallel, recombinant bacteria were expanded by inoculating the original cultures into fresh kanamycin-containing LB medium at a ratio of 1:1000. Plasmids were extracted for all five groups, C1, C2, C3, D1 and D2. Two preparations were obtained for each group and

eluted in 100 µL of EB, except for D1, for which a single preparation was obtained and eluted in 80 µL of EB. The concentration readings from the original record are listed below.

Plasmid ID	Concentration of preparation 1 (ng/µL)	Concentration of preparation 2 (ng/µL)
C1	132.05	137.10
C2	133.95	153.80
C3	149.35	156.35
D1	91.70	No second preparation
D2	103.00	91.70

8.21

Second batch of *Dunaliella salina* transformations

Cultivation of the previously transformed algae was continued, and a second batch of *D. salina* transformations was performed using osmotic shock.

Subculture of bacterial strains

DH5α cells carrying the plasmids were subcultured.

8.22

Screening across a hygromycin concentration gradient at 0.5 M NaCl

Wild-type *D. salina* was inoculated into DM medium containing 0.5 M NaCl and a range of hygromycin concentrations to identify suitable antibiotic treatment conditions. A 50 mg/mL hygromycin stock solution was used to prepare five concentrations: 100, 200, 300, 600 and 800 mg/L.

Hygromycin concentration (mg/L)	100	200	300	600	800
Volume of stock solution added (µL)	20	40	60	120	160

8.23–8.24

***Dunaliella salina* exposed to hygromycin concentration gradients at two salinities**

The 1.5 M NaCl cultures had been incubated for 4 d, whereas the 0.5 M NaCl cultures had been incubated for 24 h. Microscopy videos were recorded for cultures treated with 150 and 350 mg/L hygromycin at 1.5 M NaCl, and 800 mg/L hygromycin at 0.5 M NaCl.

Microscopy revealed motile *D. salina* cells at all hygromycin concentrations examined. Cells remained present at appreciable densities and moved rapidly over short distances. Surviving cells were also observed on the following day. These observations did not identify a hygromycin concentration that eliminated *D. salina*.

Both previous batches of transformed algae showed visible growth. The first-batch cultures had become darker green and developed some flocculent sediment during static incubation.

Microscopic examination of the sediment revealed actively motile *D. salina* cells. The bottoms of the second-batch culture tubes appeared pale green.

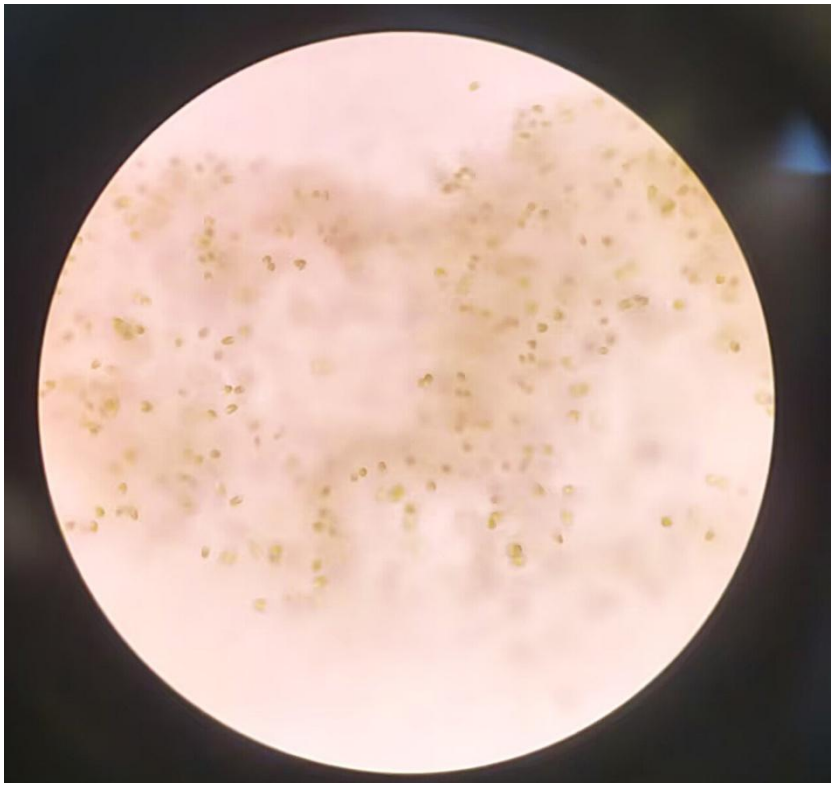


Fig. 18 | Microscopic examination of transformed *Dunaliella salina* cultures.

8.26

Extraction and analysis of algal genomic DNA

DNA was extracted from the C1 and D1 transformed algal samples using a plant genomic DNA extraction kit, including lysis, column loading, washing and elution. The recorded procedure was as follows:

1. Sample pretreatment: grind 100 mg of fresh leaves or 20–30 mg of dried tissue into a fine powder in liquid nitrogen and rapidly transfer the powder to a 2 mL microcentrifuge tube.

2. Add 400 µL of buffer GP1 to the powder and mix by vortexing. RNase A may be added beforehand if RNA removal is required.

3. Incubate in a 65 °C water bath for 10–30 min, inverting the tube several times during incubation, until the tissue is completely lysed.

4. Add 400 µL of buffer GP2 and mix thoroughly by inversion; the solution should become uniformly turbid. Centrifuge at 12,000 rpm (approximately 13,400 × g) for 5 min and transfer the supernatant to a fresh tube, avoiding the pellet.

5. Load the supernatant onto a CB3 spin column in aliquots, centrifuge at 12,000 rpm for 30 s and discard the flow-through. If the volume exceeds 700 µL, load the sample in two steps.

6. Add 500 µL of buffer GD, ensuring that ethanol has been added to the buffer, centrifuge at 12,000 rpm for 30 s and discard the flow-through.

7. Repeat the wash step once, giving two washes in total.

8. Centrifuge the empty column at 12,000 rpm for 2 min to remove residual ethanol.

9. Transfer the column to a clean tube and leave the cap open at room temperature for 2 min to dry.

10. Apply 50–100 µL of TE buffer or ddH_2O directly to the centre of the membrane without touching it. Incubate at room temperature for 2–5 min and centrifuge at 12,000 rpm for 2 min to collect the DNA.

PCR was performed using the DNA extracts as templates, with the reaction mixtures and cycling conditions shown below.

Reaction mixtures:

KOD One™ PCR Master Mix (2×)	12.5 µL
D1-CCD1-F	0.75 µL
CCD1-R	0.75 µL
ddH_2O	10 µL
Genomic DNA extract	1 µL
KOD One™ PCR Master Mix (2×)	12.5 µL
RBCS2-F	0.75 µL
CCD1-R	0.75 µL
ddH_2O	10 µL
Genomic DNA extract	1 µL

PCR cycling conditions:

Initial denaturation	98 °C	3 min	
Denaturation	98 °C	10 s	34 cycles
Annealing	65 °C	5 s	
Extension	68 °C	20 s	
Final extension	68 °C	3 min	

Electrophoresis on a 1% agarose gel revealed clear bands. The remaining PCR products were retained for sequencing.

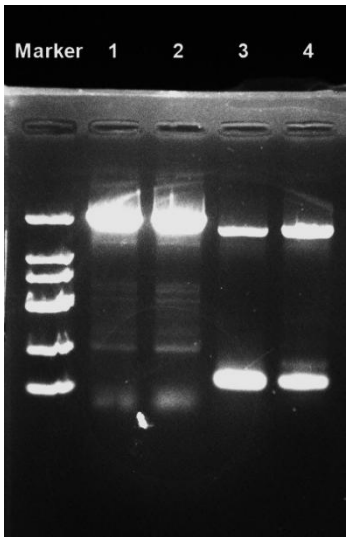


Fig. 19 | Agarose gel electrophoresis of PCR products amplified from *Dunaliella salina* genomic DNA extracts. A 100–2,000 bp DNA ladder was used.

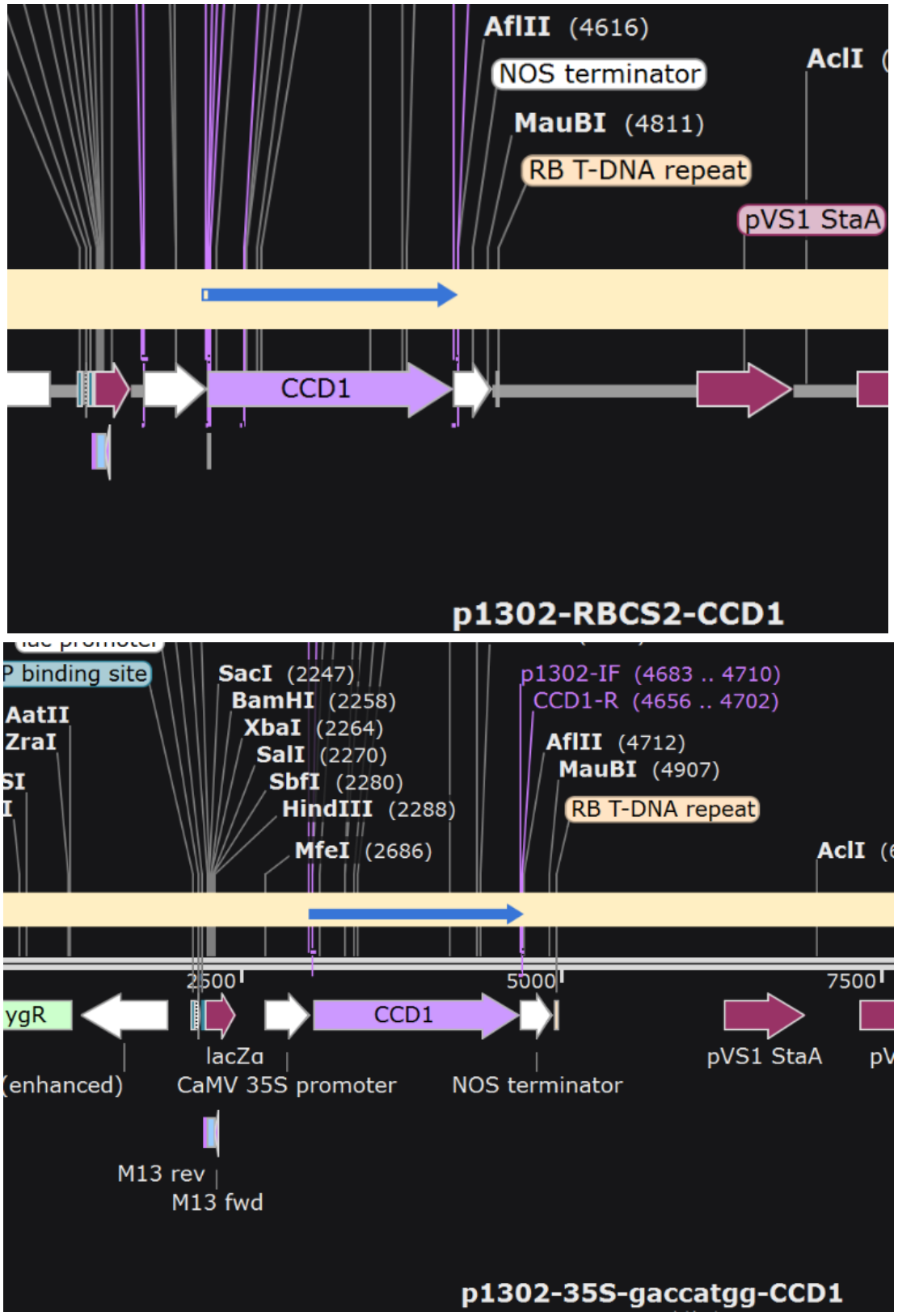


Fig. 20 | Sequencing results for *Dunaliella salina* samples. The first and second results correspond to C1 and D1, respectively.

The sequences obtained for both groups matched the expected sequences, supporting the detection of CCD1-related sequences in the genomic DNA extracts from the corresponding algal samples. These findings were interpreted as evidence of successful generation of the C1 and D1 transformed cultures.

8.27

Additional PCR products were submitted for sequencing.

Algal culture medium was prepared and sterilized for subsequent culture expansion.

8.28

Expansion of transformed algal cultures

The first batch of transformed algal cultures was transferred to conical flasks according to plasmid construct and expanded in algal medium containing 1.5 M NaCl.

Exploratory optimization of hygromycin treatment

Guided by the literature, we explored hygromycin selection in the presence of 10 mM glucose to adjust the antibiotic treatment conditions. Hygromycin concentrations of 6.25, 12.5, 75, 100 and 200 mg/L were tested.

8.30

Third round of algal transformation

Plasmids were extracted from recombinant *Escherichia coli* and used for a third round of algal transformation, yielding one treated culture each for C1, C2, C3, D1 and D2. The previously described osmotic shock transformation procedure was used.

8.31

Extraction and analysis of algal genomic DNA

DNA was extracted from the established C1, C2, C3, D1 and D2 algal cultures and analysed by PCR and agarose gel electrophoresis. The reaction mixtures and cycling conditions are listed below.

Reaction mixture for C2 and D2:

KOD One™ PCR Master Mix (2×)		12.5 µL
D1-CCD1-F		0.75 µL
CCD1-R		0.75 µL
ddH_2O		10 µL
Genomic DNA extract		1 µL

Reaction mixture for C1, C3 and D1:

KOD One™ PCR Master Mix (2×)	12.5 µL
RBCS2-F	0.75 µL
CCD1-R	0.75 µL
ddH_2O	10 µL
Genomic DNA extract	1 µL

PCR cycling conditions:

Initial denaturation	98 °C	3 min	
Denaturation	98 °C	10 s	34 cycles
Annealing	65 °C	5 s	
Extension	68 °C	20 s	
Final extension	68 °C	3 min	

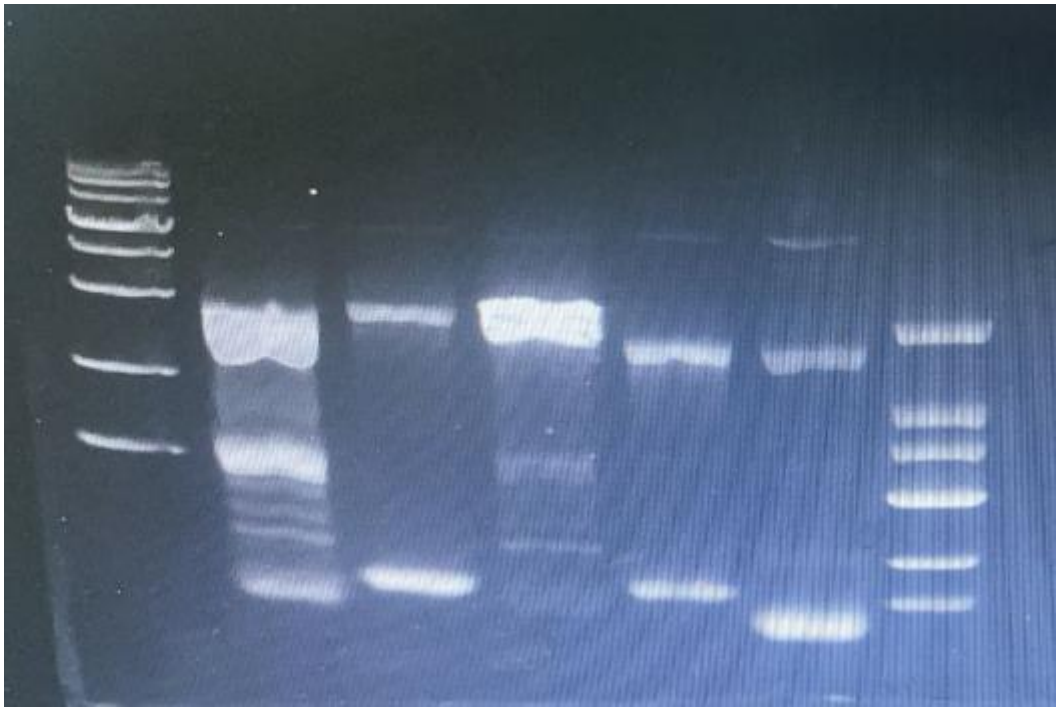


Fig. 21 | Agarose gel analysis of PCR products amplified from algal genomic DNA extracts. The first and last lanes contain 250–15,000 bp and 100–2,000 bp DNA ladders, respectively. The five intervening lanes contain C1, C2, C3, D1 and D2, in that order.

The agarose gel electrophoresis results were consistent with expectations. The DNA extracts were subsequently submitted for sequencing to further examine the presence of exogenous sequences.

9.5

Expansion of transformed algal cultures

The medium of the first batch of transformed algal cultures was replaced, and culture expansion continued.

Assessment of hygromycin treatments with glucose supplementation

The cultures treated with a hygromycin concentration gradient in the presence of glucose were examined.

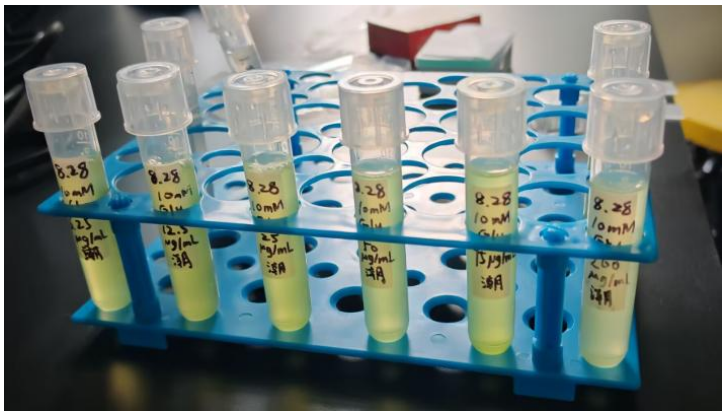


Fig. 22 | Growth assessment of algal cultures treated with glucose and a hygromycin concentration gradient.

Visual inspection showed that the culture labelled 200 µg/µL was distinctly paler green than the other cultures. Microscopy revealed that immobile cells or cells rotating in place were observed more frequently at higher hygromycin concentrations, suggesting reduced cell vitality. However, differences between adjacent treatment groups were not pronounced.

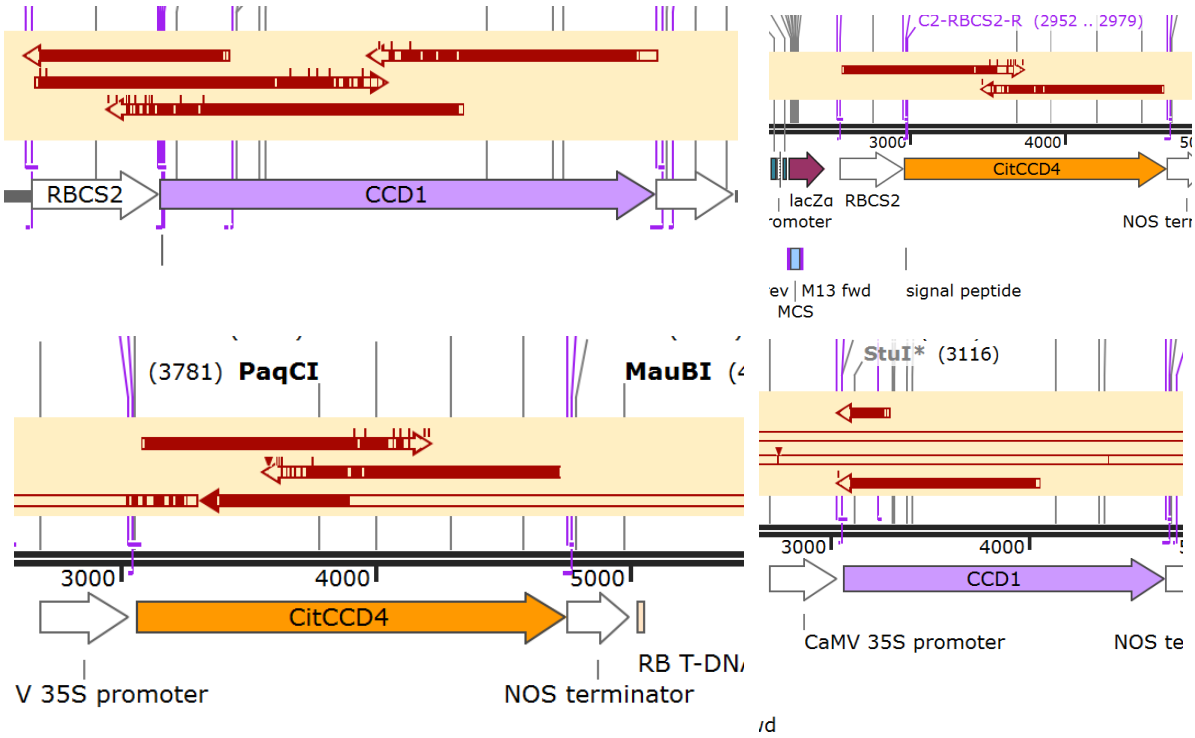
These exploratory observations suggested that glucose supplementation might influence hygromycin-mediated growth inhibition, but this experiment alone was insufficient to establish such an effect. The hygromycin concentration actually used for direct treatment was recorded as 800 µg/µL, based on earlier OD_{600} measurements and growth-curve analysis.

9.8

The C1, C2, C3, D1 and D2 algal cultures previously identified as the best-performing cultures were spread on solid DM medium containing 1.5 M NaCl and 300 mg/L hygromycin.

9.9

Sequencing results for algal genomic DNA samples



☐	0003	C1-2	DNA/CDNA	20	394	35s	No	Unidirectional	Sequencing failed No signal Re-prepare	Downloaded	2026-09-09T07:09	ab1 seq	Chromatogram
☐	0004	C2	DNA/CDNA	20	360	35s	No	Unidirectional	Sequencing failed No signal	Downloaded	2026-09-09T07:09	ab1 seq	Chromatogram
☐	0005	C3	DNA/CDNA	20	435	35s	No	Unidirectional	Sequencing failed No signal	Downloaded	2026-09-09T07:09	ab1 seq	Chromatogram
☐	0006	D1-2	DNA/CDNA	20	931	35s	No	Unidirectional	Sequencing failed No signal	Downloaded	2026-09-09T07:09	ab1 seq	Chromatogram
☐	0007	D2	DNA/CDNA	20	731	35s	No	Unidirectional	Sequencing failed No signal	Downloaded	2026-09-09T07:09	ab1 seq	Chromatogram

Fig. 23 | Sequencing results for algal genomic DNA samples.

The sequencing results from the previously submitted algal DNA samples were examined. Matches to the target gene sequences were obtained, whereas sequencing with the 35S primer yielded no signal.

Growth of the third batch of transformed algae

The third batch of transformed algal cultures also showed visible growth.

Repeat plasmid construction and transformation

The plasmid backbones were re-amplified by PCR.

The previously described reaction mixtures were used with the following cycling conditions:

Initial denaturation	98 °C	3 min	
Denaturation	98 °C	10 s	30 cycles
Annealing	60 °C	5 s	
Extension	68 °C	1 min 40 s	
Final extension	68 °C	3 min	

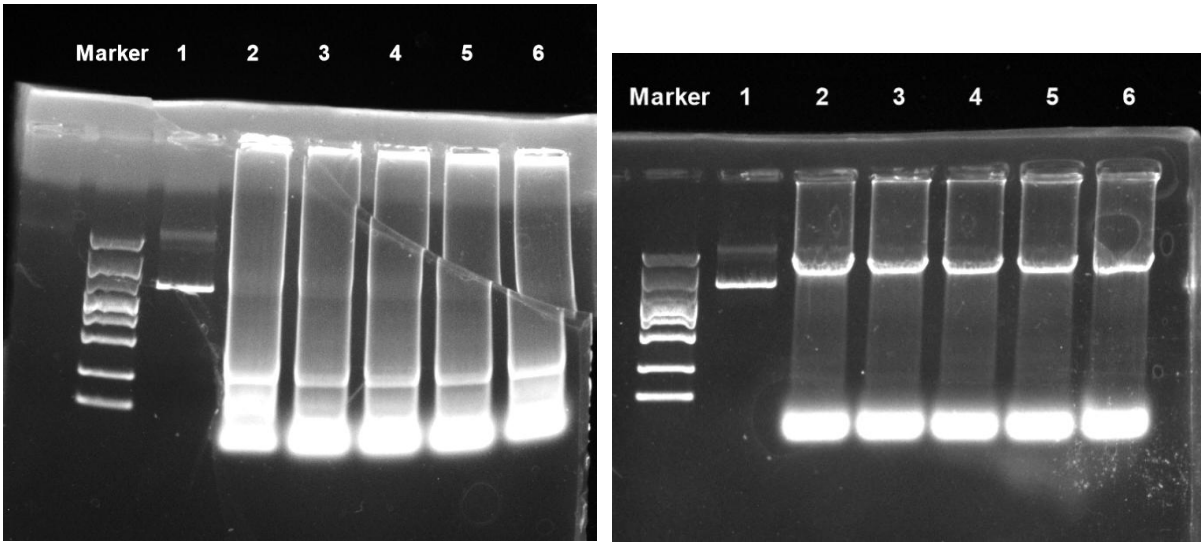


Fig. 24 | Agarose gel analysis of plasmid backbone PCR products. The images show the 35S and RBCS2 groups, respectively. Lane 1 contains the plasmid control.

Following agarose gel electrophoresis, the fragments were purified using an Omega gel extraction kit and assembled by Gibson assembly. The assembled constructs were then introduced into competent *E. coli* Stbl3 cells, which were spread on kanamycin-containing LB agar plates and incubated at 37 °C.

9.11

Expansion of single colonies

Colonies were observed on the plates for all groups except C3. Single colonies from the four groups with colony growth were selected for culture expansion.

Plasmid extraction

Plasmids were extracted from the four expanded bacterial cultures using a TIANGEN plasmid miniprep kit.

Plasmid concentrations (ng/μL):

C1	119.45
C2	177.55
D1	145.85
D2	129.25

9.12

Construction of the C3 plasmid

The recombinant C3 plasmid was reconstructed and introduced into competent *E. coli* Stbl3 cells by heat shock. The cells were spread on kanamycin-containing LB agar plates and incubated at 37 °C.

9.14

The C1, C2, D1 and D2 constructs were introduced into competent Stbl3 cells. Recombinant *E. coli* carrying C3 were also expanded to support the continued preservation and verification of all constructs.

9.15

The Stbl3 strains carrying C1, C2, C3, D1 and D2 were preserved, and plasmid extraction and colony PCR were performed. The PCR products were examined by agarose gel electrophoresis to prepare material for subsequent sequencing verification.

9.20

Colony PCR was repeated for C3 and D1. The loading order recorded in the original notebook was DNA ladder, C3 clones 2–4, DNA ladder and D1 clones 1–3.

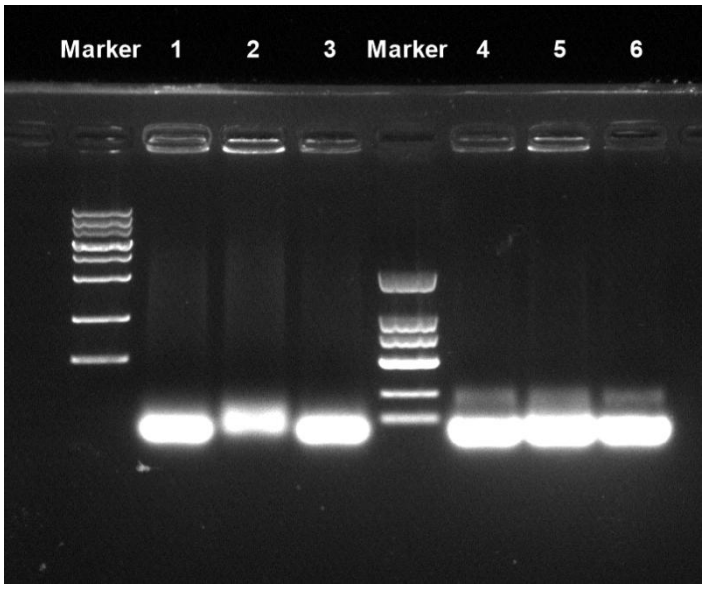


Fig. 25 | Agarose gel analysis of colony PCR products. Sample lanes 1–3 contain C3, and sample lanes 4–6 contain D1. The left and right DNA ladders span 250–15,000 bp and 100–2,000 bp, respectively.

Agarose gel electrophoresis did not identify positive clones. We suspected that the condition of the Phanta Flash PCR enzyme might have affected detection and planned to repeat the analysis using newly purchased KOD One enzyme.

Additional single colonies were selected and expanded to retain material for subsequent re-examination.

9.24

Algal transformation and cultivation

C1 DNA from sequencing samples returned on 9.19 and D2 DNA from samples returned on 9.22, together with DNA labelled c11-3, were introduced into wild-type *D. salina* using osmotic shock transformation.

Functional assessment by mass spectrometry

Algal cultures from the p1302-transformed wild-type control, C1 and C3 groups were submitted for widely targeted metabolomic analysis by ultra-performance liquid chromatography–tandem mass spectrometry (UPLC–MS/MS).

9.26

Plasmid construction workflow

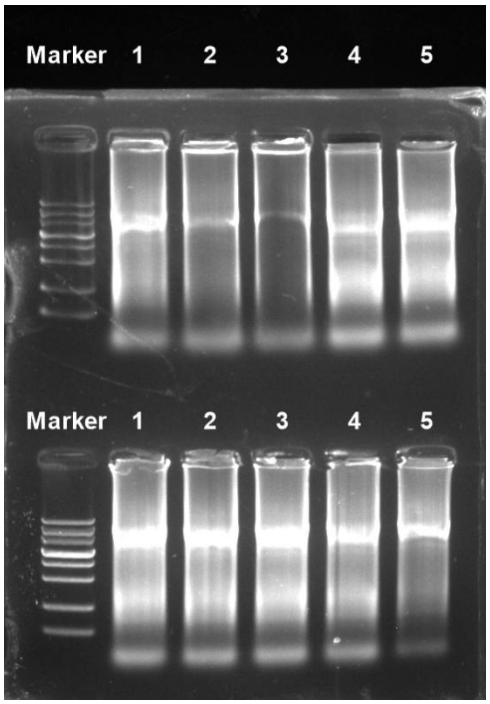


Fig. 26 | Agarose gel analysis of vector inverse PCR products. Lanes 1–3 are indicated in the original record without sample assignments. The DNA ladder spans 250–15,000 bp.

The plasmid construction workflow was repeated for each group, including vector inverse PCR, agarose gel electrophoresis, gel extraction, homology-based assembly and transformation of competent cells. Transformed cells were spread on kanamycin-containing LB agar plates and cultured.

9.30

Multiple single colonies were selected from each group for colony PCR using the previously described reaction mixtures and cycling conditions.

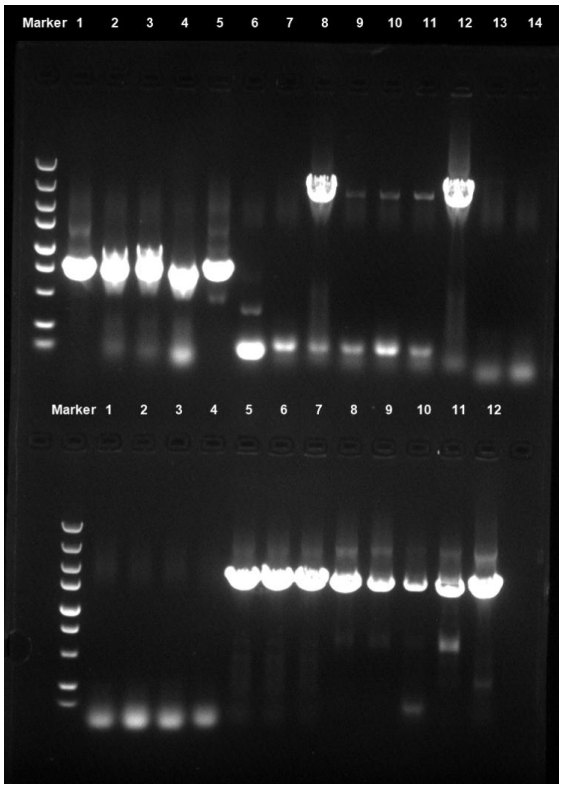


Fig. 27 | Agarose gel analysis of colony PCR products. The DNA ladder spans 100–5,000 bp.

10.1

Positive colonies were expanded, plasmids were extracted, and algal transformation was performed using the osmotic shock method.

10.9

Genomic DNA was extracted from the different batches of algal cultures and analysed by PCR and agarose gel electrophoresis.

The electrophoresis results were unsatisfactory.

10.10–10.20

RNA was extracted from *Dunaliella salina* using a Vazyme RNA extraction kit and analysed by RT-qPCR.

Extracts from each of the two engineered *D. salina* strains carrying the biosynthetic genes were analysed by thin-layer chromatography (TLC) alongside extracts from the wild-type (WT) control. No β-citraurin reference standard was available. The TLC results for β-ionone were inconclusive owing to its high volatility, and the β-ionone standard eventually evaporated completely.

The results of the mass spectrometry-based metabolomic analysis became available on 10.13.

These results are described in detail in the Results and Engineering sections.